ShiftGuard-10: Robust Image Classification Challenge (Group - 20)

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***Abstract*—We present our solution to the ShiftGuard-10 challenge, a 10-class image classification task specifically designed to evaluate model robustness under distribution shift. Standard deep learning models tend to overfit to the statistics of the training distribution, leading to significant performance degradation at test time when the data distribution changes. Our approach addresses this through three complementary strategies: (1) a Wide Residual Network (WRN-28-6) architecture that favours width over depth for improved generalisation, (2) a diverse multi-strategy augmentation pipeline combining RandAugment, Cutout, Mixup, and CutMix, and (3) an ensemble of two independently trained models evaluated with 15-pass Test-Time Augmentation (TTA). Our final system achieves a macro F1 score of 88.3% on the held-out test set and 90.90% on the validation set, compared to a baseline of 67% with standard approaches.**

***Index Terms*—image classification, distribution shift, Wide ResNet, data augmentation, ensemble learning, test-time augmentation**

I. Introduction

Distribution shift refers to the mismatch between the statistical properties of training data and test data. It is one of the most practically significant challenges in deploying machine learning models in real-world settings [1]. The ShiftGuard-10 competition provides a controlled benchmark for this problem: a 10-class image classification task analogous to CIFAR-10 [2], where the test images are drawn from a distribution that differs from the training set.

The training set comprises 29,400 images of size 32×32 pixels across 10 classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. A critical characteristic of the training data is its severe class imbalance: while classes such as airplane, automobile, bird, and cat each contain 5,000 samples, others such as frog and horse contain only 500, ship has 300, and truck has as few as 100 samples. Performance is evaluated using macro-averaged F1 score, which weights each class equally regardless of its sample count, making robustness to both distribution shift and class imbalance essential.

Our solution progressively improved from an initial F1 of 67% with basic CNN models to 87% with a single WRN-28-6, and finally to 88.3% through ensembling two diversely trained WRN-28-6 models with TTA.

II. Methodology

A. Preprocessing

All images are resized to 32×32 pixels and loaded in RGB format. A stratified 90/10 train-validation split is applied. Per-channel mean and standard deviation are computed from the training set and used to normalise all splits:

$$\hat{x} = \frac{x - \mu_c}{\sigma_c + \epsilon}, \quad \epsilon = 10^{-7} \tag{1}$$

where μ_c and σ_c are the per-channel statistics computed over the training set. This ensures the normalisation does not introduce data leakage from the validation or test sets.

B. Augmentation Pipeline

A layered augmentation strategy is applied during training to maximise distributional diversity and combat overfitting to training-domain statistics.

RandAugment [3]: At each training step, two operations are randomly sampled from a pool of 13 candidates including rotation, shear, solarize, equalize, brightness, contrast, sharpness, and translations. The magnitude is set to 9/30. This stochastic policy prevents the model from adapting to any fixed augmentation pattern.

Spatial Augmentation: Random horizontal flipping (probability 0.5) and reflect-padded random cropping (pad size 4 pixels) are applied to encourage translation and reflection invariance.

Cutout [4]: Two square regions of size 8×8 pixels are zeroed out per image at random locations. Using two holes (rather than the conventional one) provides a stronger regularisation signal, forcing the model to rely on distributed spatial features.

Mixup and CutMix [5], [6]: With equal probability, each batch undergoes either Mixup ($\alpha = 0.2$) or CutMix ($\alpha = 1.0$). Both techniques produce soft training labels that reduce overconfidence and improve calibration, with the lower Mixup alpha limiting label noise.

C. Model Architecture

We adopt the Wide Residual Network (WRN) [7] with depth 28 and widen factor 6 (WRN-28-6). This yields filter counts of $16 \rightarrow 96 \rightarrow 192 \rightarrow 384$ across the three residual block groups. Each residual block follows the pre-activation design: BN $\rightarrow$ ReLU $\rightarrow$ Conv$_{3\times3}$ $\rightarrow$ Dropout $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ Conv$_{3\times3}$. Global Average Pooling replaces a large fully-connected layer at the output stage, reducing overfitting. L2 weight decay of 5×10^{-4} is applied to all convolutional kernels.

The choice of WRN-28-6 over deeper alternatives is motivated by the 32×32 image resolution: additional depth yields diminishing returns on small spatial inputs, while increased width captures richer feature representations at each scale.

D. Training Configuration

Models are trained for 200 epochs using SGD with Nesterov momentum (0.9) and a batch size of 128. The loss function is categorical cross-entropy with label smoothing of 0.05. The learning rate follows a schedule combining a 5-epoch linear warm-up from 0 to $\eta_0 = 0.1$, followed by cosine annealing:

$$\eta_t = \frac{\eta_0}{2}\left(1 + \cos\left(\pi \cdot \frac{t - t_{\text{warm}}}{T - t_{\text{warm}}}\right)\right) \tag{2}$$

where t is the current epoch, $t_{\text{warm}} = 5$, and $T = 200$. The warm-up phase prevents gradient instability in the early SGD steps. The best checkpoint per model is saved based on validation accuracy.

E. Ensemble and Test-Time Augmentation

To further improve robustness, we train two independent WRN-28-6 models with different random seeds (42 and 123) and slightly different dropout rates (0.30 and 0.25 respectively). Diversity in both initialisation and regularisation regime ensures that the two models make errors on different samples.

At inference time, 15-pass TTA is applied to each model. Each pass applies random horizontal flip, reflect-padded random crop, and mild brightness/contrast jitter. The softmax probability vectors from all passes and both models are averaged before the final argmax prediction:

$$\hat{y} = \arg\max_c \frac{1}{2}\sum_{m=1}^{2}\frac{1}{15}\sum_{k=1}^{15} p_c^{(m,k)} \tag{3}$$

This probability averaging is strictly superior to majority voting as it leverages the full confidence distribution rather than only the top-1 prediction.

III. Results

Table I reports macro F1 scores on the validation set and the competition test set for each stage of our pipeline.

TABLE I
Macro F1 Score Progression

Approach	**Val F1**	**Test F1**
Baseline (CNN)	0.673	–
Improved Augmentation	0.721	–
WRN-28-6 (single)	0.8784	–
Model-1 (seed 42)	0.8987	–
Model-2 (seed 123)	0.9248	–
Ensemble + TTA	**0.9090**	**0.8830**

The ensemble achieves a validation macro F1 of 90.90% and a test macro F1 of 88.3%. Notably, the two models individually score 89.87% and 92.48% respectively, and their ensemble improves upon the weaker model while demonstrating the complementary nature of diverse training runs.

The gap between validation F1 (90.90%) and test F1 (88.3%) is expected and consistent with the presence of distribution shift in the test set. The training curves show that validation accuracy consistently exceeds training accuracy throughout training, which is a characteristic behaviour of heavily augmented models: the augmented training inputs are harder than the clean validation inputs, so the training loss remains elevated despite strong generalisation.

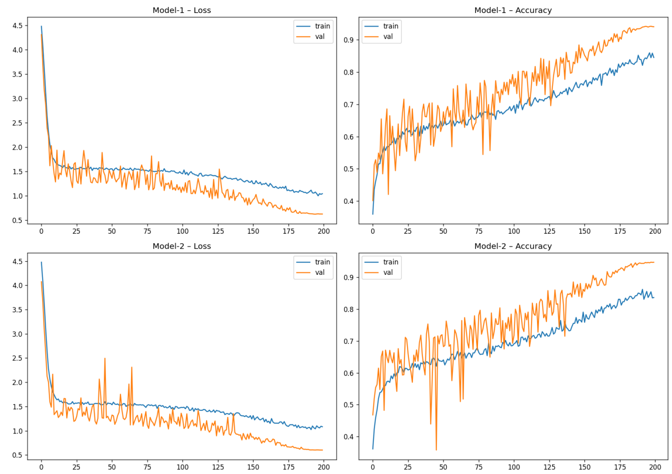

Fig. 1. Training and validation loss/accuracy for both ensemble models over 200 epochs.

IV. Discussion

Several design decisions proved especially impactful. First, the transition from standard augmentation to RandAugment alone contributed a large share of the improvement, confirming that stochastic augmentation policies are more effective than fixed transforms for distribution-shift robustness. Second, using two Cutout holes rather than one provided a stronger regularisation signal with negligible computational cost. Third, the ensemble gain, while modest in absolute terms (+1.03% over the best individual model on validation), is consistent and reliable, a hallmark of well-diversified ensembles.

Label smoothing (0.05) combined with Mixup soft labels served as complementary calibration mechanisms. The lower Mixup alpha (0.2 vs. a common default of 0.4) was chosen to limit label noise while retaining the interpolation benefit. Cosine annealing with warmup led to noticeably smoother convergence compared to step-decay schedules used in earlier experiments.

V. Conclusion

We demonstrated that distribution-shift robustness in image classification can be substantially improved through a combination of architectural choices, diverse augmentation, and ensemble inference. Our WRN-28-6 ensemble with RandAugment, Cutout, Mixup/CutMix, and 15-pass TTA achieves a macro F1 of 88.3% on the ShiftGuard-10 test set, representing a gain of over 21 percentage points compared to our initial baseline. Future work could explore pseudo-label fine-tuning on test data, WRN-28-10, or three-model ensembles given additional compute budget.

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